

```
1
2 options nodate nonumber;
3 ods graphics on;
4
5
6 proc import datafile="/home/u64435956/MedendorpSpencerStat448/acad_dig_new1 (1).csv"
7     out=students
8     dbms=csv
9     replace;
10    guessingrows=max;
11 run;
12
13 title "Preview of Imported Data";
14 proc print data=students(obs=15);
15 run;
16
17 title "Variable Names and Data Types";
18 proc contents data=students;
19 run;
20
21
22 data students_clean;
23     set students;
24
25     drop student_id;
26
27     label
28         age = "Age"
29         gender = "Gender"
30         study_hours_per_day = "Study Hours Per Day"
31         smartphone_usage_hours = "Smartphone Usage Hours"
32         social_media_hours = "Social Media Hours"
33         gaming_hours = "Gaming Hours"
34         streaming_hours = "Streaming Hours"
35         sleep_hours = "Sleep Hours"
36         exercise_hours = "Exercise Hours"
37         class_attendance_percent = "Class Attendance Percent"
38         assignment_completion_percent = "Assignment Completion Percent"
39         caffeine_intake_cups = "Caffeine Intake Cups"
40         mental_health_status = "Mental Health Status"
41         parent_education_level = "Parent Education Level"
42         internet_quality = "Internet Quality"
43         motivation_level = "Motivation Level"
44         final_exam_score = "Final Exam Score";
45 run;
46
47
48 title "Missing Value Check for Numeric Variables";
49
50 proc means data=students_clean n nmiss mean min max maxdec=2;
51     var age
52         study_hours_per_day
53         smartphone_usage_hours
54         social_media_hours
55         gaming_hours
56         streaming_hours
57         sleep_hours
```

```
58     exercise_hours
59     class_attendance_percent
60     assignment_completion_percent
61     caffeine_intake_cups
62     motivation_level
63     final_exam_score;
64 run;
65
66 title "Categorical Level Counts";
67
68 .....
69 proc freq data=students_clean;
70     tables gender
71             mental_health_status
72             parent_education_level
73             internet_quality
74             / missing;
75 run;
76
77 title "Descriptive Statistics for Continuous Variables";
78
79 .....
80 proc means data=students_clean n mean median std min max maxdec=2;
81     var age
82         study_hours_per_day
83         smartphone_usage_hours
84         social_media_hours
85         gaming_hours
86         streaming_hours
87         sleep_hours
88         exercise_hours
89         class_attendance_percent
90         assignment_completion_percent
91         caffeine_intake_cups
92         motivation_level
93         final_exam_score;
94 run;
95
96 title "Frequency Tables for Categorical Variables";
97
98 .....
99 proc freq data=students_clean;
100     tables gender
101             mental_health_status
102             parent_education_level
103             internet_quality
104             / missing;
105 run;
106
107 title "Distribution of Final Exam Scores";
108
109 .....
110 proc sgplot data=students_clean;
111     histogram final_exam_score;
112     density final_exam_score;
113     xaxis label="Final Exam Score";
114     yaxis label="Frequency";
115 run;
```

```
116
117 title "Correlations with Final Exam Score";
118
119 proc corr data=students_clean pearson spearman;
120     var final_exam_score;
121     with age
122         study_hours_per_day
123         smartphone_usage_hours
124         social_media_hours
125         gaming_hours
126         streaming_hours
127         sleep_hours
128         exercise_hours
129         class_attendance_percent
130         assignment_completion_percent
131         caffeine_intake_cups
132         motivation_level;
133 run;
134
135
136 title "T-Test: Final Exam Score by Gender";
137
138 proc ttest data=students_clean;
139     class gender;
140     var final_exam_score;
141 run;
142
143
144 title "ANOVA: Final Exam Score by Mental Health Status";
145
146 proc glm data=students_clean;
147     class mental_health_status;
148     model final_exam_score = mental_health_status;
149 run;
150 quit;
151
152
153 title "ANOVA: Final Exam Score by Parent Education Level";
154
155 proc glm data=students_clean;
156     class parent_education_level;
157     model final_exam_score = parent_education_level;
158 run;
159 quit;
160
161
162 title "ANOVA: Final Exam Score by Internet Quality";
163
164 proc glm data=students_clean;
165     class internet_quality;
166     model final_exam_score = internet_quality;
167 run;
168 quit;
169
170
171 title "Variable Selection Using Stepwise AIC";
172
173
```

```
174 proc glmselect data=students_clean;
175     class gender
176         mental_health_status
177         parent_education_level
178         internet_quality;
179
180     model final_exam_score =
181         age
182         gender
183         study_hours_per_day
184         smartphone_usage_hours
185         social_media_hours
186         gaming_hours
187         streaming_hours
188         sleep_hours
189         exercise_hours
190         class_attendance_percent
191         assignment_completion_percent
192         caffeine_intake_cups
193         mental_health_status
194         parent_education_level
195         internet_quality
196         motivation_level
197     / selection=stepwise(select=aic choose=aic)
198       stats=all;
199 run;
200
201
202
203 title "Final Selected Multiple Regression Model";
204
205 .....
206 proc glm data=students_clean;
207     class mental_health_status
208         parent_education_level;
209
210     model final_exam_score =
211         study_hours_per_day
212         smartphone_usage_hours
213         gaming_hours
214         sleep_hours
215         exercise_hours
216         class_attendance_percent
217         assignment_completion_percent
218         caffeine_intake_cups
219         mental_health_status
220         parent_education_level
221         motivation_level
222     / solution clparm;
223
224     output out=final_model_output
225         p=predicted_score
226         r=residual;
227 run;
228 quit;
229
230 .....
231 data final_model_output;
```

```
232     set final_model_output;
233     squared_error = residual**2;
234 run;
235
236
237 title "Original Model RMSE";
238
239 proc means data=final_model_output mean maxdec=4;
240     var squared_error;
241     output out=rmse_original mean=mse;
242 run;
243
244
245 data rmse_original;
246     set rmse_original;
247     rmse = sqrt(mse);
248 run;
249
250
251 proc print data=rmse_original;
252     var mse rmse;
253 run;
254
255
256 title "Original Model Residual Plot";
257
258
259 proc sgplot data=final_model_output;
260     scatter x=predicted_score y=residual;
261     refline 0 / axis=y;
262     xaxis label="Predicted Final Exam Score";
263     yaxis label="Residual";
264 run;
265
266
267 title "Original Model Predicted vs Actual Plot";
268
269 proc sgplot data=final_model_output;
270     scatter x=predicted_score y=final_exam_score;
271     reg x=predicted_score y=final_exam_score;
272     xaxis label="Predicted Final Exam Score";
273     yaxis label="Actual Final Exam Score";
274 run;
275
276
277 title "Original Model Q-Q Plot of Residuals";
278
279 proc univariate data=final_model_output normal;
280     var residual;
281     qqplot residual / normal(mu=est sigma=est);
282 run;
283
284
285 title "VIF Check for Numeric Predictors";
286
287 proc reg data=students_clean;
288     model final_exam_score =
289         study_hours_per_day
```

```
290     smartphone_usage_hours
291     gaming_hours
292     sleep_hours
293     exercise_hours
294     class_attendance_percent
295     assignment_completion_percent
296     caffeine_intake_cups
297     motivation_level
298     / vif tol;
299 run;
300 quit;
301
302
303
304 title "Cook's Distance Influence Diagnostics";
305
306 -----
307 proc reg data=students_clean;
308     model final_exam_score =
309         study_hours_per_day
310         smartphone_usage_hours
311         gaming_hours
312         sleep_hours
313         exercise_hours
314         class_attendance_percent
315         assignment_completion_percent
316         caffeine_intake_cups
317         motivation_level
318         / influence;
319
320     output out=cooks_output
321            cookd=cooks_d
322            p=predicted_score
323            r=residual;
324 run;
325 quit;
326
327 -----
328 data cooks_flagged;
329     set cooks_output;
330
331     cooks_cutoff = 4 / 500;
332
333     if cookd > cooks_cutoff then influential = 1;
334     else influential = 0;
335 run;
336
337
338 title "Number of Influential Observations";
339
340 -----
341 proc freq data=cooks_flagged;
342     tables influential;
343 run;
344
345 title "Influential Observations Based on Cook's Distance";
346
347 -----
348 proc print data=cooks_flagged;
```

```
348     where influential = 1;
349     var final_exam_score
350         predicted_score
351         residual
352         cooks_d
353         study_hours_per_day
354         smartphone_usage_hours
355         gaming_hours
356         sleep_hours
357         exercise_hours
358         class_attendance_percent
359         assignment_completion_percent
360         caffeine_intake_cups
361         motivation_level;
362 run;
363
364
365 -----
366 data students_no_influence;
367     set cooks_flagged;
368     if influential = 0;
369 run;
370
371 title "Sample Size After Removing Influential Observations";
372
373 -----
374 proc means data=students_no_influence n;
375     var final_exam_score;
376 run;
377
378 title "Final Model After Removing Influential Observations";
379
380 -----
381 proc glm data=students_no_influence;
382     class mental_health_status
383           parent_education_level;
384
385     model final_exam_score =
386         study_hours_per_day
387         smartphone_usage_hours
388         gaming_hours
389         sleep_hours
390         exercise_hours
391         class_attendance_percent
392         assignment_completion_percent
393         caffeine_intake_cups
394         mental_health_status
395         parent_education_level
396         motivation_level
397         / solution clparm;
398
399     output out=final_model_no_influence
400           p=predicted_score
401           r=residual;
402 run;
403 quit;
404
405
```

```
406 data final_model_no_influence;
407     set final_model_no_influence;
408     squared_error = residual**2;
409 run;
410
411
412 title "RMSE After Removing Influential Observations";
413
414 proc means data=final_model_no_influence mean maxdec=4;
415     var squared_error;
416     output out=rmse_no_influence mean=mse;
417 run;
418
419
420 data rmse_no_influence;
421     set rmse_no_influence;
422     rmse = sqrt(mse);
423 run;
424
425
426 proc print data=rmse_no_influence;
427     var mse rmse;
428 run;
429
430
431 title "Residual Plot After Removing Influential Observations";
432
433
434 proc sgplot data=final_model_no_influence;
435     scatter x=predicted_score y=residual;
436     refline 0 / axis=y;
437     xaxis label="Predicted Final Exam Score";
438     yaxis label="Residual";
439 run;
440
441
442 title "Predicted vs Actual After Removing Influential Observations";
443
444
445 proc sgplot data=final_model_no_influence;
446     scatter x=predicted_score y=final_exam_score;
447     reg x=predicted_score y=final_exam_score;
448     xaxis label="Predicted Final Exam Score";
449     yaxis label="Actual Final Exam Score";
450 run;
451
452
453 data rmse_original_clean;
454     set rmse_original;
455     model = "Original Model";
456     keep model mse rmse;
457 run;
458
459
460 data rmse_no_influence_clean;
461     set rmse_no_influence;
462     model = "Cook's Distance Model";
463     keep model mse rmse;
464 run;
465
466
467 data rmse_comparison;
```

```
464      set rmse_original_clean rmse_no_influence_clean;
```

```
465  run;
```

```
466
```

```
467
```